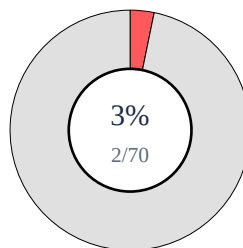


SCRIBIFY AI — STUDENT EVALUATION REPORT

Student ID: Bhavanya_256



Total: 2/70
Percentage: 3.2%
Attempted: 5/27

Part A | Q1

Score: 0.25/1

Question: What is the significance of the discovery of Coelacanth in 1938?

Student Answer: In 1938, 1958, Coelacanth was caught in a lake in South Africa after it was thought to have become extinct. They are also called lobefins & they evolved to become fish & first amphibians that lived in waters & also on land. We have no specimens of these left today.

- **Correct:** The student correctly identifies the year of the discovery (1938) and the key aspect that Coelacanths were believed to be extinct before their rediscovery.
- **Wrong / Missing:** The answer incorrectly states the Coelacanth was caught in a lake (it's a marine fish) and includes the extraneous year 1958. Most importantly, the statement that 'We have no specimens of these left today' is factually incorrect and directly contradicts the significance of its discovery as a 'living fossil' that is still alive. The provided textbook emphasizes that Coelacanths 'are still alive today'. The mention of lobefins and evolution to amphibians, while related to Coelacanths, is not highlighted as the primary significance of the 1938 discovery in the provided text.
- **Improve:** Focus on the precise significance mentioned in the text: that the discovery proved a species thought to be extinct was actually still alive, making it a 'living fossil'. Ensure all factual claims align with the current scientific understanding and the provided text.
- **Ref:** Summary section

Part A | Q2

Score: 1.0/1

Question: What are interferons and their role in the immune system?

Student Answer: The interferons are a part of cytokine barrier. They are released by virus infected cells to protect the normal cells from further viral infection. They are also used in the treatment of cancer.

- **Correct:** You correctly identified interferons as part of the cytokine barrier, secreted by virus-infected cells, and stated their role in protecting non-infected cells from further viral infection.
- **Wrong / Missing:** The provided textbook context does not mention interferons being used in the treatment of cancer.
- **Improve:** Ensure your answers are directly supported by the provided textbook context.
- **Ref:** Cytokine barriers

Part A | Q3

Score: 0.0/1

Question: Define dominance and incomplete dominance in genetics.

Student Answer: Dominance Incomplete dominance.

- **Correct:**
- **Wrong / Missing:** The student did not provide any definition for either dominance or incomplete dominance. The answer merely restates the terms from the question.
- **Improve:** Define dominance as the phenomenon where one allele masks the expression of another in a heterozygote, and incomplete dominance as when the heterozygous phenotype is an intermediate blend of the two homozygous parental phenotypes.
- **Ref:** Section 5.2.1 and 5.2.2.1

Part A | Q4

Score: 1.0/1

Question: Where do the signals of parturition originate in mammals?

Student Answer: The signals of parturition originate from fully developed foetus and the placenta of the female.

- **Correct:** You correctly identified that the signals for parturition originate from the fully developed fetus and the placenta.
- **Wrong / Missing:** Nothing is wrong or missing in your answer.
- **Improve:** Your answer is accurate and complete. Keep up the great work!
- **Ref:** Parturition and Lactation section

Part A | Q5

Score: 0.0/1

Question: What are cry I Ac and cryAb genes?

Student Answer: cry I Ac and cry II Ab (Gene name written in Italics)

- **Correct:** The provided textbook context does not contain information regarding cry I Ac and cryAb genes, so a correct explanation cannot be derived from it.
- **Wrong / Missing:** The student's answer merely restates the gene names, with a slight alteration (cry II Ab instead of cryAb). It does not explain **what** these genes are, which is what the question asks. The necessary information to answer this question is not present in the provided textbook context.
- **Improve:** To answer a 'What are...' question, you need to provide a description, definition, or function of the item in question, rather than just repeating its name. Ensure your answer is based on the provided learning material.
- **Ref:** Not found in provided text.

Part B | Q6

Score: 0.0/2

Question: Explain Thalassemia as an autosomal recessive trait. Describe the genetic basis. How can parents be carriers for Thalassemia? What genes are involved?

Student Answer:

- **Correct:**
- **Wrong / Missing:** Answer not provided.
- **Improve:** Attempt the question with definition, key points, and example.
- **Ref:** Chunk 46; Chunk 73; Chunk 148

Part B | Q7

Score: 0.0/2

Question: Describe the production and use of Polyblend in road construction.

Student Answer:

- **Correct:**
 - **Wrong / Missing:** Answer not provided.
 - **Improve:** Attempt the question with definition, key points, and example.
 - **Ref:** Chunk 200; Chunk 206; Chunk 176
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Part B | Q8

Score: 0.0/2

Question: List surgery, radiotherapy, and chemotherapy as cancer treatments. Briefly describe each.

Student Answer:

- **Correct:**
- **Wrong / Missing:** Answer not provided.
- **Improve:** Attempt the question with definition, key points, and example.
- **Ref:** Chunk 117; Chunk 116; Chunk 101

Part B | Q9

Score: 0.0/2

Question: Define Net Primary Productivity (NPP) and Gross Primary Productivity (GPP).

Student Answer:

- **Correct:**
- **Wrong / Missing:** Answer not provided.
- **Improve:** Attempt the question with definition, key points, and example.
- **Ref:** Chunk 176; Chunk 201; Chunk 183

Part B | Q10

Score: 0.0/2

Question: State the competitive exclusion principle and resource partitioning in ecology.

Student Answer:

- **Correct:**
- **Wrong / Missing:** Answer not provided.
- **Improve:** Attempt the question with definition, key points, and example.
- **Ref:** Chunk 170; Chunk 183; Chunk 187

Part B | Q11

Score: 0.0/2

Question: Describe the function of sporogenous tissue and tapetum in plants.

Student Answer:

- **Correct:**
- **Wrong / Missing:** Answer not provided.
- **Improve:** Attempt the question with definition, key points, and example.
- **Ref:** Chunk 21; Chunk 163; Chunk 10

Part B | Q12

Score: 0.0/2

Question: Discuss the ability to freeze and transport semen for artificial insemination.

Student Answer:

- **Correct:**
- **Wrong / Missing:** Answer not provided.
- **Improve:** Attempt the question with definition, key points, and example.
- **Ref:** Chunk 41; Chunk 125; Chunk 38

Part C | Q13

Score: 0.0/3

Question: Name three microbes (e.g., *Penicillium notatum*, *Monascus purpureus*, *Trichoderma polysporum*) and their major applications in medicine or industry.

Student Answer:

- **Correct:**
- **Wrong / Missing:** Answer not provided.
- **Improve:** Attempt the question with definition, key points, and example.
- **Ref:** Chunk 133; Chunk 135; Chunk 110

Part C | Q14

Score: 0.0/3

Question: Describe the steps of Polymerase Chain Reaction (PCR) and the role of *Thermus aquaticus*.

Student Answer:

■ **Correct:**

■ **Wrong / Missing:** Answer not provided.

■ **Improve:** Attempt the question with definition, key points, and example.

■ **Ref:** Chunk 68; Chunk 174; Chunk 69

Part C | Q15

Score: 0.0/3

Question: Explain Meselson-Stahl experiment and semi-conservative replication of DNA.

Student Answer:

- **Correct:**
- **Wrong / Missing:** Answer not provided.
- **Improve:** Attempt the question with definition, key points, and example.
- **Ref:** Chunk 68; Chunk 145; Chunk 80

Part C | Q16

Score: 0.0/3

Question: Discuss the process of emasculation and bagging for artificial hybridization in plants.

Student Answer:

- **Correct:**
- **Wrong / Missing:** Answer not provided.
- **Improve:** Attempt the question with definition, key points, and example.
- **Ref:** Chunk 24; Chunk 129; Chunk 23

Part C | Q17

Score: 0.0/3

Question: a) Draw a neat labeled diagram showing the development of ovarian follicles and corpus luteum. b) Mention the role of FSH and LH in this process.

Student Answer:

- **Correct:**
- **Wrong / Missing:** Answer not provided.
- **Improve:** Attempt the question with definition, key points, and example.
- **Ref:** Chunk 30; Chunk 32; Chunk 29

Part C | Q18

Score: 0.0/3

Question: Explain plant cell totipotency and its application in tissue culture.

Student Answer:

- **Correct:**
- **Wrong / Missing:** Answer not provided.
- **Improve:** Attempt the question with definition, key points, and example.
- **Ref:** Chunk 21; Chunk 163; Chunk 149

Part C | Q19

Score: 0.0/3

Question: Using the example of the peppered moth, describe natural selection in response to pollution.

Student Answer:

- **Correct:**
- **Wrong / Missing:** Answer not provided.
- **Improve:** Attempt the question with definition, key points, and example.
- **Ref:** Chunk 86; Chunk 138; Chunk 139

Part C | Q20

Score: 0.0/3

Question: Explain the technique and application of mutation breeding in crop improvement.

Student Answer:

- **Correct:**
- **Wrong / Missing:** Answer not provided.
- **Improve:** Attempt the question with definition, key points, and example.
- **Ref:** Chunk 129; Chunk 127; Chunk 151

Part C | Q21

Score: 0.0/3

Question: Describe the structure and function of restriction endonucleases and sticky ends for DNA fingerprinting.

Student Answer:

■ **Correct:**

■ **Wrong / Missing:** Answer not provided.

■ **Improve:** Attempt the question with definition, key points, and example.

■ **Ref:** Chunk 144; Chunk 145; Chunk 70

Part C | Q22

Score: 0.0/3

Question: What is Phenylketonuria? State its symptoms and inheritance. Describe Klinefelter's Syndrome and its effects.

Student Answer:

- **Correct:**
- **Wrong / Missing:** Answer not provided.
- **Improve:** Attempt the question with definition, key points, and example.
- **Ref:** Chunk 59; Chunk 61; Chunk 57

Part C | Q23

Score: 0.0/3

Question: Define carrying capacity and logistic growth in populations. How do predators regulate population equilibrium?

Student Answer:

- **Correct:**
- **Wrong / Missing:** Answer not provided.
- **Improve:** Attempt the question with definition, key points, and example.
- **Ref:** Chunk 166; Chunk 164; Chunk 180

Part C | Q24

Score: 0.0/3

Question: What is DNA fingerprinting? List its steps and mention its use in paternity disputes

Student Answer:

- **Correct:**
- **Wrong / Missing:** Answer not provided.
- **Improve:** Attempt the question with definition, key points, and example.
- **Ref:** Chunk 79; Chunk 68; Chunk 145

Part D | Q25

Score: 0.0/5

Question: Define and differentiate natality, immigration, mortality, and emigration in population studies. Explain direct and indirect methods of population estimation.

Student Answer:

- **Correct:**
- **Wrong / Missing:** Answer not provided.
- **Improve:** Attempt the question with definition, key points, and example.
- **Ref:** Chunk 164; Chunk 166; Chunk 165

Part D | Q26

Score: 0.0/5

Question: Describe the role of zona pellucida in fertilization. What is the function of acrosome in sperm during fertilization?

Student Answer:

- **Correct:**
- **Wrong / Missing:** Answer not provided.
- **Improve:** Attempt the question with definition, key points, and example.
- **Ref:** Chunk 26; Chunk 9; Chunk 6

Part D | Q27

Score: 0.0/5

Question: Explain transcription and translation differences between prokaryotes and eukaryotes.

Student Answer:

- **Correct:**
- **Wrong / Missing:** Answer not provided.
- **Improve:** Attempt the question with definition, key points, and example.
- **Ref:** Chunk 71; Chunk 76; Chunk 81

